

Graduate School of Science and Technology Doctoral Thesis Abstract

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Title	Improving Electricity Theft Detection for Smart Homes: Insights from Real and Synthetic Attack Scenarios		

Abstract:

Efforts to improve the security of electricity grids include both physical measures, such as detecting interference, and digital security methods, such as encryption. However, these measures alone are insufficient for addressing the full range of cyberattacks. For example, digital electricity meters, which are essential to new infrastructure, are vulnerable to software and hardware issues, and the digitization of these meters reintroduces concerns about electricity theft, which can now manifest as IT-related problems. To address these concerns, contemporary approaches that use data analytics, machine learning (ML), and predictive techniques are required. The rise in sophisticated statistical methods, particularly those involving machine learning, has led to an interest in developing models and algorithms, e.g., for smart homes, that can interpret smart meter data to quickly identify signs of tampering.

Smart home appliances, which are becoming increasingly integral to modern households, are also vulnerable to electricity theft. This can have a significant impact on both utility providers and consumers, as these appliances often connect to the internet and transmit usage data to utility providers for billing and monitoring. This data transmission can be intercepted or manipulated, leading to false readings and unauthorized electricity use. The ability to remotely control appliances also presents a risk of unauthorized access, which can lead to the misuse or manipulation of consumption data. To address these vulnerabilities, emerging technologies such as advanced ML algorithms and enhanced encryption methods are being developed to detect anomalies in usage patterns and secure data transmission. These technologies can help reduce the vulnerability of smart home appliances to electricity theft and improve the overall security of the electricity grid.

This dissertation presents novel approaches to electricity theft detection (ETD) by introducing various classification algorithms to detect anomalies in non-intrusive appliance load monitoring (NIALM) or disaggregated smart meter networks. Our proposed framework utilizes ML knowledge-based synthetic attack data (KBSAD) to train an attack classifier. These data consist of benign attack patterns that serve as the foundation for generating synthetic and simulated attacks that closely mirror real-world scenarios. The framework was validated using the Almanac of Minutely Power dataset version 2 (AMPds2), which contains fine-grained time-series data from a smart home. We preprocessed the data for binary classification to evaluate our synthetic attack model using real attack data. The Extreme Gradient Boosting algorithm performed best with an average area under curve (AUC) score of 98.74% and 98.69% for detecting and classifying anomalies in real and simulated attacks, respectively. These methods outperformed legacy unsupervised methods (LUM). By integrating the KBSAD, our approach eliminates the need for extensive data collection for real attacks and seamlessly combines synthetic attacks with genuine consumption readings, representing a significant advancement in the field of smart-home electricity theft detection.